(b) *Materials and Storage*

- Piling and stacking–Can material slip easily ?
- Materials protruding out of racks, bins, benches, machines etc.

(c) *Building*

- Windows clean and unbroken.
- Painting and upkeep.
- Door jambs clean.
- Fire extinguishers and sprinklers clear.

(d) *Floors*

- Slippery, wet or oily.
- Badly worn or rutted.
- Garbage, dirt or debris.
- Loose materials.

(e) *Stairways and Aisles*

- Clear and unblocked
- Well lighted.

(f) *Employee facilities*

- Drinking taps clean.
- Toilets and locker rooms clean.
- Soap and towels available.

(g) *Other Aspects*

- Lamps and lamp reflectors clean.
- Bulletin boards and safety signs clean.
- Protective equipment and clothing clean and in good condition.
- Electrical motors clean.
- Ventilation unobstructed.

20.9. PERSONNEL (LABOUR) WELFARE

Definition and Concept

- Employee services are provided under a number of titles in industry. Sometimes, it is referred as
  Benefit Programmes,
  Personnel (employee, labour) welfare, or as
  Hidden payroll.

Perhaps employee services are generally described as a part of *Fringe Benefits.*

- Fringe benefits include those elements of compensation other than wages which are significant when initially considering employment and in the ongoing evaluation of one's welfare on the job.

Fringe benefits may be of

(*i*) Monetary nature, *e.g.*, Retirement benefits, Insurance benefits, and Investment plans, etc.

(*ii*) Non-monetary nature, *e.g.*, Position title, Good office, Automobile, and Good parking space etc.

Importance (and Objectives) of

Personnel welfare or employee service and benefit programmes are important because they :

(1) Make the plant personnel a healthier, sounder-thinking and more forward-looking group.

(2) Make the employee a group of citizens better able to carry on the productive processes.

(3) Contribute to the maintenance of employee morale and loyalty.

(4) Maintain an employee's favourable attitude towards his work and work environment.

(5) Serve to attract and keep a work force in competition with other organizations.

(6) Serve to maintain some degree of peace with the organized labour union.

(7) Fulfil social, recreational, and cultural needs of the employees and at the same time make their life easier.

(8) Reduce labour turn-over and absenteeism.

(9) Promote good public relations.

(10) Encourage employees and promote goodwill and cordial relations between employers and employees, which ultimately result in more production, better product quality and increased profits.

Welfare (and Services) Methods or Measures

They may be categorised into three classes, namely

(1) Economic.

(2) Recreational.

(3) Facilitative.

1. *Economic*

(*a*) Insurance (including group insurance)

(*b*) Retirement and pension plans.

(*c*) Health and accident services.

(*d*) Credit unions[1].

(*e*) Paid holidays.

(*f*) Profit sharing.

2. *Recreational*

(*a*) Sports.

(*b*) Social get-togethers.

(*c*) Special interest groups such as dramatics, athletic programmes, flying and particular hobbies.

3. *Facilitative*

(*a*) Housing.

(*b*) Transport.

(*c*) Educational facilities and library services.

(*d*) Medical services (including first-aid, hospitalization, sick leave, etc.).

(*e*) Canteens, cafeterias and lunch wagons (Eating facilities).

(*f*) Company (cheap) stores.

(*g*) Discounts on purchases of company products.

(*h*) Rest-rooms and locker-rooms.

(*i*) Legal and financial counselling.

The above listed facilities are self-explanatory.

1. A credit union is an organized group of people who pool their money and agree to make loans to one another at the time of need.

he is at study.

- Position Rotation. Rotating an executive from one position to another broadens his background in the business.
- Special Projects. A special assignment, *e.g.*, "to develop a system of dust collection in the foundry," is highly useful and flexible training device.
- Committee Assignments. Unlike special projects, committee assignments are regularly constituted. Committee assignments very well impart the necessary general background.
- Selective Reading, *e.g.*, going through Business Magazines, Trade Journals, etc.

(*ii*) *Off-the-Job training and development which includes*

- Attending special courses conducted at colleges and universities.
- Role Playing. It involves constructing artificially a conflict situation in which the trainee is given a strategic position to play.

Role playing increases the trainee's skill in dealing with other people (*e.g.*, Sales Training).

- Sensitivity Training. It develops executive's awareness and sensitivity to behavioural patterns of oneself and other people.
- Simulation. Trainees are asked to make decisions about production, cost, inventories, sales etc., for a simulated firm.
- Conference Training (discussed earlier).
- Attending Special Meetings of one or two days duration in various fields, *e.g.*, Personnel, Production or Marketing Management, etc.

20.6. SAFETY ENGINEERING

Safety in Industry

- The modern safety movement started around 1912 with the First Cooperative Safety Congress and the organization of the National Safety Council in U.S.A.

From 1912 to the present time, remarkable advances have been made in reducing the rate and severity of accidents.

- The importance of industrial safety was realized because every year millions of industrial accidents occur which result in either death or in temporary and permanent disablement of the employees and involve a good amount of cost such as resulting from wasted manhours, machine hours etc.
- In 1952 in U.S.A., fifteen thousand workers were killed in industrial accidents, 2,000,000 were injured and the total cost of these accidents was about $ 2,900,000,000.
- Loss of lives and accidents costs gradually led to the formation of Factories act, Office, Shops and Railway Premises Act etc.
- The requirement for consideration of safety by management as part of its responsibility arises primarily from these Acts.
- Safety begins on the drawing board when in the original design of tools or workplace layout, accident hazard may be built in or eliminated.
- *Safety results*

  (*i*) from safe plant, processes, and operations, and

  (*ii*) by educating and training workers and supervisors regarding safe practices on the shop floor.
- In an industry, safety may be considered from the mechanical side (equipment, tools etc.) or from legel angles of workmen's compensation or even as a matter of training in and motivation towards safe work practices for workers (especially newly recruited ones).

Need for Safety

Safety in industry helps,

(*i*) Increasing rate of production.

(*ii*) Reducing production cost.

(*iii*) Reducing damage to equipment and machinery.

(*iv*) Preventing premature death of talented workers who are an asset to the society.

(*v*) Preventing needless pain and suffering to its emplyees.

Organization for Safety

- In a small concern each shop supervisor may be made responsible for safety in his shop.
- Each shop supervisor may report to top executive as regards safety matters.
- Since the shop supervisor has its main job to turn out production, he may treat safety as a secondary aspect.
- For this reason sometimes the safety function is taken care of by personnel officer or general foreman.
- With the growth in the size of the industry and depending upon the hazardousness of processes/operations, a full fledged safety department may be created with the safety Director/Manager as its chief executive and a number of persons under him at different levels.

The safety Director/Manager may be given a line position or staff position depending upon the conditions in the industry.

- Sometimes the responsibility for safety rests on a safety committee.

Safety Committee

- A safety committee may consist of executives, supervisors, and shop floor workers.
- Thus the lower level employees get a channel of communication on safety matters direct to executive level.
- It was observed that those organizations which made safety committees had lower record of accidents than those without safety committees.
- Safety committees aid in developing safety consciousness as well as it is a policy making body on such safety matters as come before it.
- The Safety Manager/executive requires a degree of firmness and ready discrimination to exclude personal and union matters in which safety is merely the pretext for their airing.
- The safety executive should guard jealously the responsibilities of management and supervision.
- Lastly, to get maximum out of a safety committee

(*i*) It should be assigned specific problems and duties such as planning safety rules, publicizing them etc.

(*ii*) Its members should be asked to go on the shop floor and watch what is being done about it (*i.e.*, the safety).

(*iii*) It should be asked to report periodically as what improvements have been made and what more can be done.

Safety Programmes

- A safety programme tends to discover when, where and why accidents occur.
- A safety programme aims at reducing accidents and the losses associated with them.
- A safety programme begins with the assumption that most work-connected accidents can be prevented.

- A safety programme does not have an end ; rather it is a continuous process to achieve adequate safety.
- A safety programme tries to reduce the influence of personal and environmental factors that cause accidents.
- A safety programme involves providing, safety equipments and special training to employees.
- A safety programme is composed of one or more of the following elements :
  - *(i)* Support by top management.
  - *(ii)* Appointing a Safety Director.
  - *(iii)* Engineering a safe plant, processes and operations.
  - *(iv)* Educating all employees to work safely.
  - *(v)* Studying and analysing the accidents to prevent their occurrence in future.
  - *(vi)* Holding safety contests, safety weeks etc., and giving incentives/prizes to departments having least number of accidents.
  - *(vii)* Enforcing safety rules.
- A safety programme includes mainly four E's (as explained above also) :
  - *(i)* *Engineering i.e.*, safety at the design and equipment installation stage.
  - *(ii)* *Education* of employees in safe practices.
  - *(iii)* *Enlistment.* It concerns the attitude of employees and management toward the programme and its purpose. It is necessary to arouse the interest of employees in accident prevention and safety-consciousness.
  - *(iv)* *Enforcement, i.e.*, to enforce adherence to safety rules and safe pratices.

Safety Instructions and Training

- This is essential for educating the employees to think, act and work safely so that the number of accidents can be minimized.
- Safety training/education gives knowledge about safe (and unsafe) mechanical conditions, personal practices and of the remedial measures.
- Safety training involves :
  - *(i)* Induction and orientation of new recruits to safety rules and practices.
  - *(ii)* Explaining safety function, during, on the job training.
  - *(iii)* Efforts made by the first level supervisors.
  - *(iv)* Formulating employees safety committees.
  - *(v)* Holding of special employee safety meetings.
  - *(vi)* Displaying charts, posters, films etc., to emphasize the need to act safely.

Educating Employees to Develop Safety Consciousness

- A worker will usually accept the use of a safety measure if he is convinced of its necessity.

Therefore, suitable measures should be adopted to increase the awareness of a need for safety in the environment of work.

- Some such measures to develop safety consciousness among workers/employees are as follows :
  - *(i)* Display of safety posters and films to remind workers of particular hazards/accidents.
  - *(ii)* Providing simple and convenient safety devices.
  - *(iii)* Providing allowance (in the standard time) to the worker for setting, removing and replacing any necessary safety devices.
  - *(iv)* Ask the employee from the first day he starts work to adopt safety measures because a

worker who has commenced work and has become familiar with it would never feel the need for safety measures at a later date.

(*v*) Hold safety competitions and award prizes to the winners.

(*vi*) Give due respect and recognition to safe workers and create in employees a feeling of pride in safe work.

(*vii*) Elaborate on the safety theme until all the employees are safety-conscious.

(*viii*) Hold regular safety meetings. They stimulate ideas and workers get more safety conscious as the time of meeting approaches near.

(*ix*) Lay out work areas to reflect safety considerations.

(*x*) Mail information and literature pertaining to safety at the homes of all employees.

(*xi*) Report safety activities to all employees.

(*xii*) Welcome all safety suggestions.

(*xiii*) Cross-mark all accident areas.

(*xiv*) Conduct safety training lectures periodically.

20.7. ACCIDENTS

– An industrial accident may be defined as an event, detrimental to the health of man, suddenly occurring and originating from external sources, and which is associated with the performance of a paid job, accompanied by an injury, followed by disability or even death. An accident may happen to any employee under certain circumstances.

Economic Aspects (Cost) of Accidents

– An accident can be very costly to the injured employee as well as to the employer of the concern.

– There are definite costs associated with the accident, *e.g., direct and measurable costs* and *indirect, i.e.*, somewhat intangible but nevertheless real *costs*.

Direct Costs of an Accident. They associate :

(*i*) Compensation insurance, including Payment, and Overhead costs.

(*ii*) Uncompensated wage losses of the injured employee.

(*iii*) Cost of medical care and hospitalization.

Indirect Costs of an Accident. They associate :

(*i*) Costs of damage to equipment, materials and plant.

(*ii*) Costs of wages paid for time lost by workers not injured.

(*iii*) Costs of wages paid to the injured worker.

(*iv*) Costs of safety engineers, supervisors and staff in investigating, recording and reporting of accidents and its causes.

(*v*) Costs of replacing the injured employee.

(*vi*) Cost of lowered production by the substitute worker.

(*vii*) Cost of delays in production due to accident.

(*viii*) Cost of reduction in efficiency of the injured worker when he joins the concern after getting recovered.

And lastly the influence of accident on the morale of employees.

Example 20.1. *Cost of an accident.* A foundary worker got burns on his foot while pouring molten metal from the ladle into the mold.